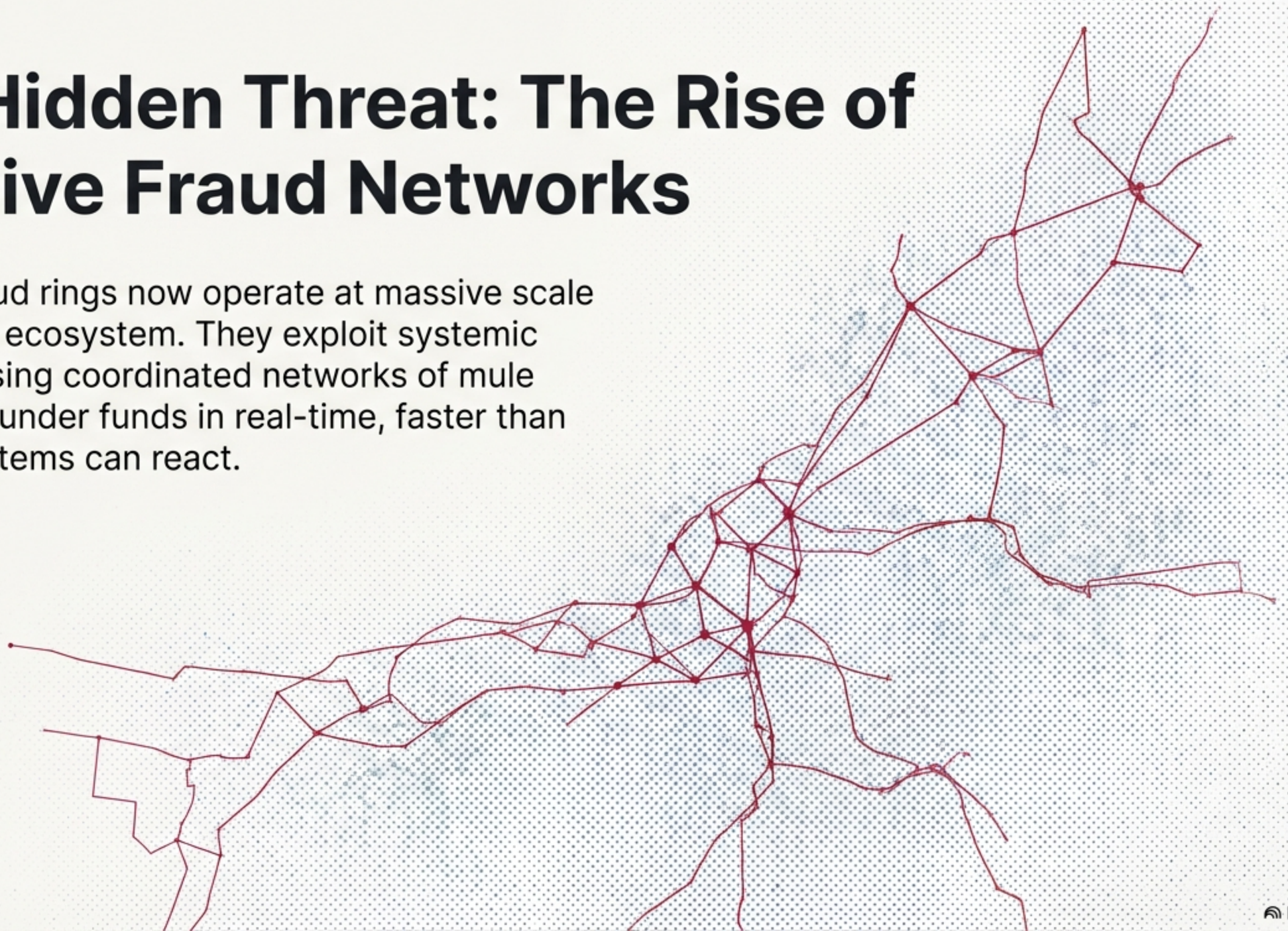
# UPI’s Hidden Threat: The Rise of Collusive Fraud Networks

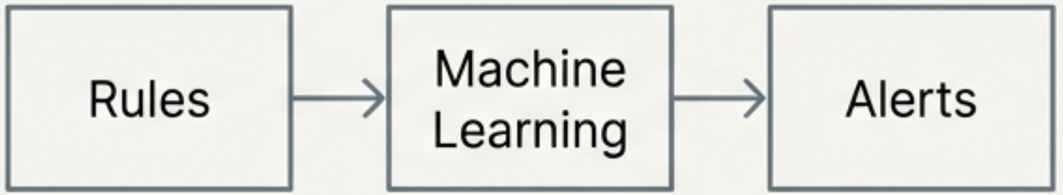
Organized fraud rings now operate at massive scale within the UPI ecosystem. They exploit systemic blind spots, using coordinated networks of mule accounts to launder funds in real-time, faster than traditional systems can react.



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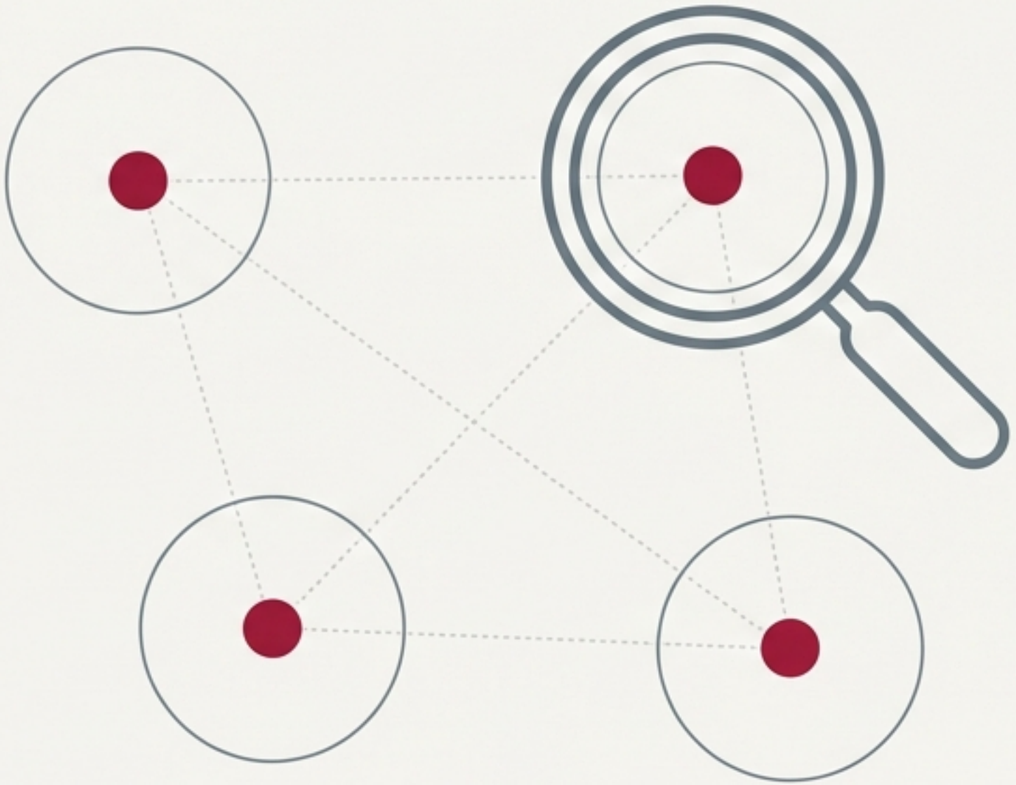
# Today’s Fraud Systems See Accounts, Not Networks

## The Current Paradigm: Reactive & Account-Focused




- ✕ Focuses on individual accounts, not coordinated behavior.
- ✕ Relies on batch analysis, missing real-time threats.
- ✕ Detects mules only *after* significant activity.
- ✕ **Crucially, they miss early-stage collusion.**

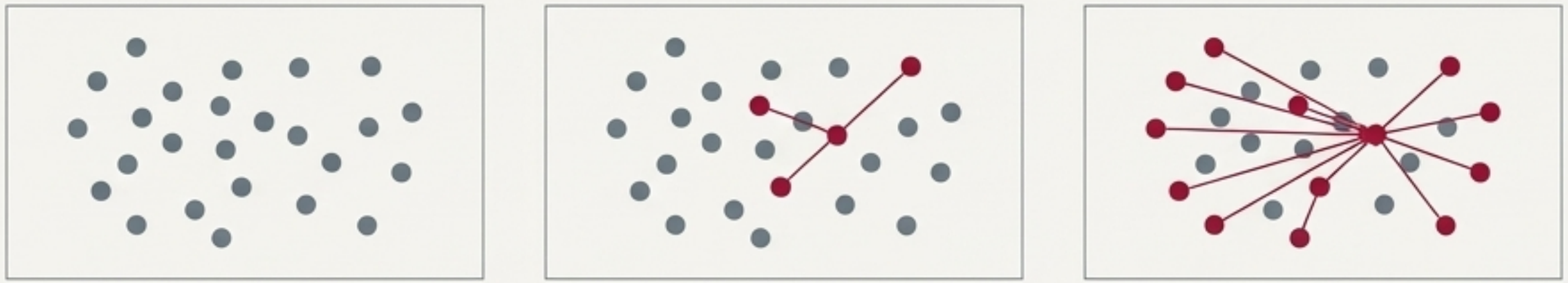
## The Result: A Systemic Blind Spot



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# Our Approach is Fundamentally Different

**“We detect fraud as an EMERGING GRAPH PHENOMENON, not as isolated events.”**



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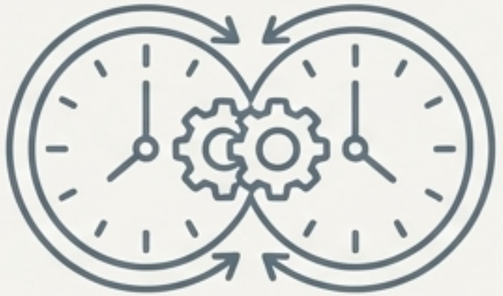
# Four Pillars of Real-Time Fraud Intelligence

Our platform is built on four novel layers of analysis that work in concert to identify and neutralize collusive threats at their inception.



## The Living Graph

Real-time network formation.



## The Collusion Signature

Identifying synchronized behavior.



## The Root of Identity

A device-first view of fraud.



## The AI Investigator

Agentic analysis and explanation.

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# Pillar 1: We Analyze the Living Transaction Graph

## Streaming Graph Engine

Every UPI transaction updates a live, in-memory graph of all entities: UPI IDs, devices, bank accounts, and IPs.

This allows us to analyze the network as it forms, not hours or days later.

## Mule Account Early-Birth Detection

We detect mule accounts at the moment of their creation by identifying suspicious “trust inheritance”—when a new account immediately transacts with known high-risk hubs, bypassing the slow growth of a legitimate user.



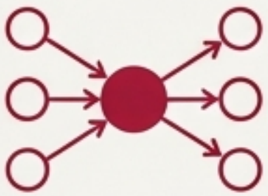
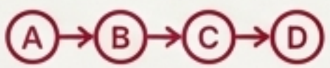
> 💡 **Innovation:** Fraud is detected **while the graph is still forming**, not after losses are incurred.

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# Pillar 2: We Identify the Signature of Collusion

## Graph Motifs

We detect known structural patterns of fraud rings beyond simple cycles:

-  Fan-in → Fan-out (Collection & Distribution)
- 

Daisy Chains (Layering)
-  Burst-based ‘Wash’ Flows

## Temporal & Amount Synchrony

Fraud rings are clock-driven. We detect when multiple, supposedly independent accounts transact:

- Within **±2 seconds** of each other.
- Using identical amounts (e.g., ₹9,999) or algorithmic splits.
- From the same device fingerprint.



> 💡 **Innovation**: Collusion is **synchronized behavior**, not just money flow.

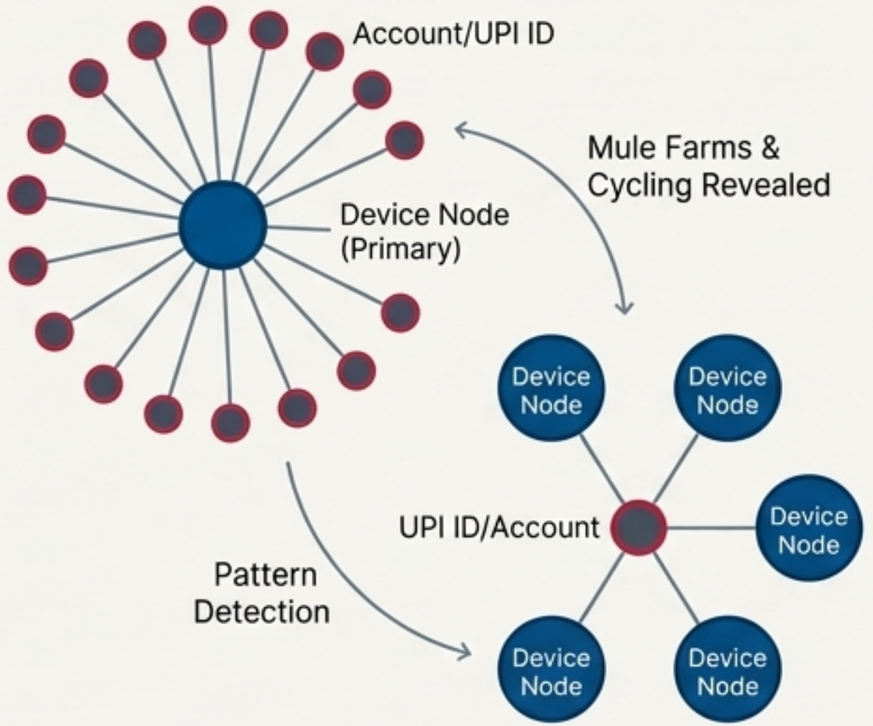
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# Pillar 3: We Anchor Identity to the Device, Not the Account

## The Device-Identity Shadow Graph

We build a parallel graph where the device is the primary node and accounts are secondary attributes. This instantly reveals mule farms farms and account cycling.

- We detect patterns like: One device → many new UPI IDs, or one UPI ID → rapidly rotating devices.




## Technology

This is achieved using a privacy-preserving device fingerprint derived from browser/app entropy, network jitter, and OS signals.

All identifiers are hashed and tokenized to be RBI-friendly.

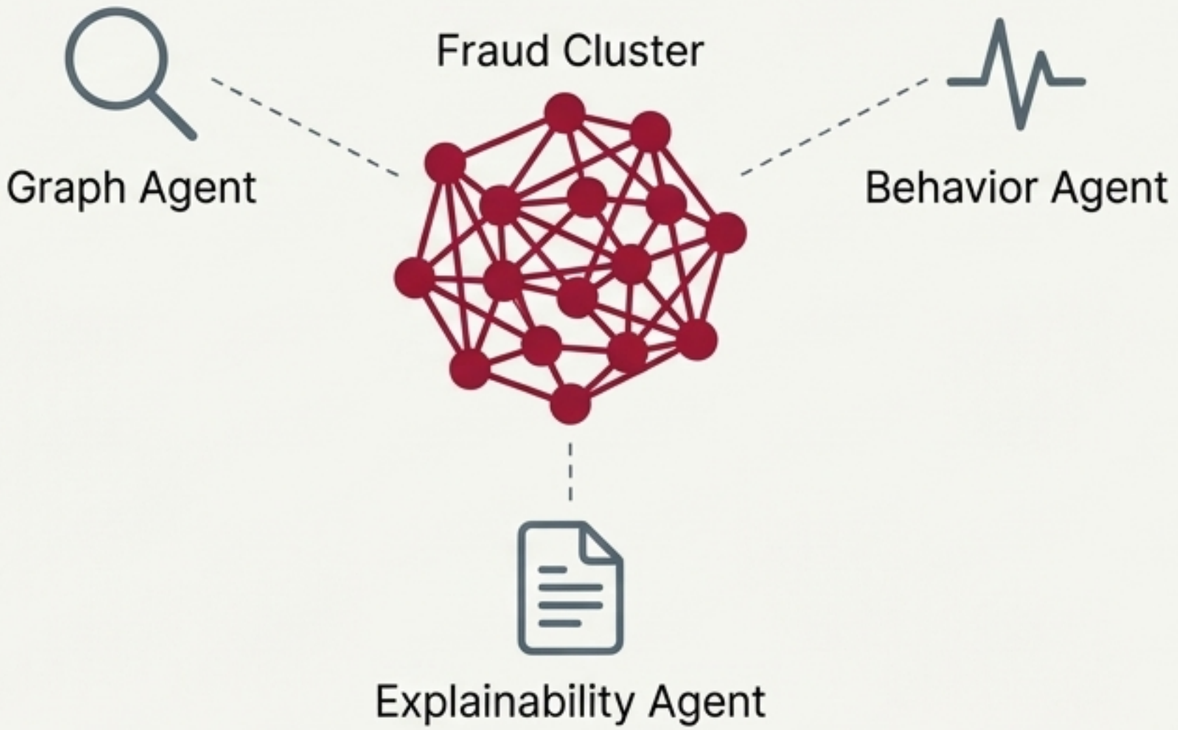
> 💡 **Key Innovation:** The device becomes the **root of the fraud graph**, providing a stable anchor that criminals cannot easily change.

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# Pillar 4: An Agentic AI That Investigates for You

UPI-NETRA employs a team of AI agents that autonomously investigate suspicious activity.

- **Graph Agent:** Finds and expands suspicious subgraphs.
- **Behavior Agent:** Flags abnormal user journeys and timing.
- **Explainability Agent:** Generates investigator-ready narratives.




```
> "This UPI ID is part of a 14-node ring sharing 3 devices, with synchronized ₹9,999 transfers occurring within 4 seconds."
```

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# A Real-Time, Production-Ready Risk API

A single, low-latency API call provides a comprehensive risk assessment during the transaction flow.

## /upi/risk-score

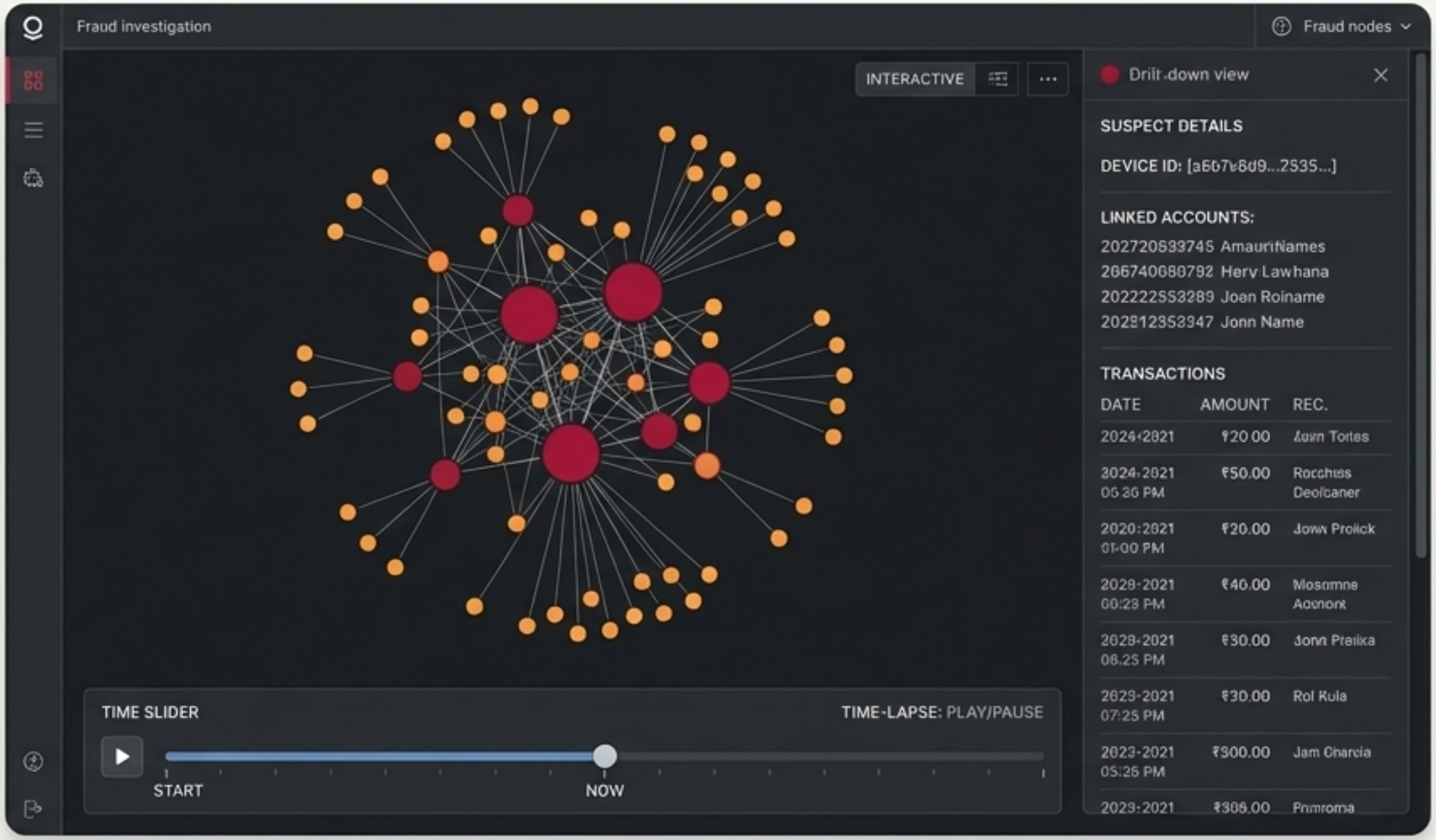

```
{
  "risk_score": 0.92,
  "reason_codes": [
    "NEW_ACCOUNT_HIGH_VELOCITY",
    "SHARED_DEVICE_WITH_FLAGGED_NODES",
    "TEMPORAL_COLLUSION_DETECTED"
  ],
  "ring_probability": 0.88,
  "mule_stage": "ACTIVE",
  "recommended_action": "STEP_UP_AUTH"
}
```


**Response time: <50ms**

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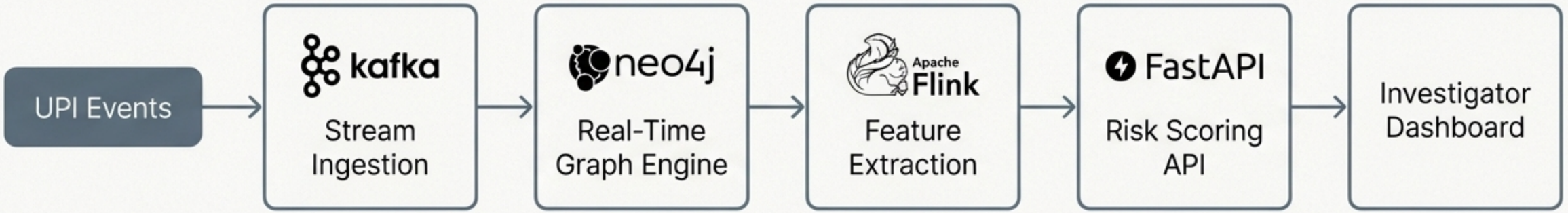
# An Investigator Dashboard Built for Intelligence, Not Just Alerts

- Interactive Fraud Cluster Exploration
- Time-Lapse Fraud Evolution
- Device-to-Account Pivoting
- Confidence Score Overlays




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# An Architecture Designed Designed for India-Scale




Our architecture uses an industry-standard, microservices-based approach built on proven open-source technologies, ensuring scalability to millions of transactions per second and compatibility with existing banking infrastructure.

**Tech Stack Highlights**

kafka neo4j Apache Flink FastAPI FingerprintJS PyTorch Geometric AnythingLLM

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# Measuring What Matters: Proactive Detection

Beyond standard metrics, we introduce a new KPI that directly measures the core value of our proactive approach.

| Standard KPIs | Our Innovation Metric |
| --- | --- |
| • Precision / Recall<br>• False Positive Reduction vs. Baseline<br>• Detection Latency (<50ms) | **Fraud Ring Discovery Time**<br>The time elapsed from the first collusive transaction within a new network to its successful detection by the system. |

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# Our Core Identity

Not just another fraud model.

We are building **Fraud Intelligence Infrastructure.**

UPI-NETRA provides not just scores, but a living, explorable map of emerging threats, empowering investigators with the context needed to dismantle entire criminal networks, not just block individual accounts.

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# UPI-NETRA

## India-scale. Investigator-ready.
## The future UPI security layer.

A real-time fraud intelligence platform that detects mule accounts and collusive UPI fraud rings by watching how networks, devices, and behaviors evolve — not just transactions.

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